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Physical constraints and biological regulations underlie universal osmoresponses

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This manuscript develops a theoretical model of osmotic pressure adaptation in microbes by osmolyte production and wall synthesis. The prediction of a rapid increase in growth rate on osmotic shock is experimentally validated using fission yeast. By using phenomenological rules rather than detailed molecular mechanisms, the model can potentially apply to a wide range of microbes, providing **important** insights that would be of interest to the wider community studying the regulation of cell size and mechanics. The level of coarse-graining and the assumptions and limitations of the model have been well described, providing a **convincing** foundation for making predictions. However, further experimental work on the validity of the core assumptions across a range of microbial organisms is needed to assess the universality of the model.

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Abstract

Microorganisms constantly transition between environments with dramatically different external osmolarities. However, theories of microbial osmoresponse integrating physical constraints and biological regulations are lacking. Here, we propose such a theory, utilizing the separation of timescales for passive responses and active regulations. We demonstrate that regulations of osmolyte production and cell-wall synthesis assist cells in coping with intracellular crowding effects and adapting to a broad range of external osmolarity. Furthermore, we predict a threshold value above which cells cannot grow, ubiquitous across bacteria and yeast. Intriguingly, the theory predicts a dramatic speedup of cell growth after an abrupt decrease in external osmolarity due to cell-wall synthesis regulation. Our theory rationalizes the unusually fast growth observed in fission yeast after an oscillatory osmotic perturbation, and the predicted growth rate peaks match quantitatively with experimental measurements. Our study reveals the physical basis of osmoresponse, yielding farreaching implications for microbial physiology.

Microbes constantly transition between environments with dramatically different osmolarities, a hallmark of microbial life [1–4]. One of the most essential features of walled microbial cells is the turgor pressure — the elastic stress stretching the cell wall due to osmotic imbalance. Upon a hypoosmotic shock (i.e., a sudden decrease of the external osmolarity), the turgor pressure increases immediately due to the sudden water influx. To relax the turgor pressure, the cell up-regulates the cell-wall synthesis rate, adds more materials to the peptidoglycan network, and eventually adapts to the lower external osmolarity [5]. Upon a hyperosmotic shock (i.e., a sudden increase of the external osmolarity), the cell volume of a microbial cell shrinks within milliseconds due to water efflux, leading to a decreased turgor pressure [6, 7]. To increase the internal osmotic pressure, microorganisms increase their intracellular solute pool by amassing osmolyte molecules (i.e., osmoregulation), e.g., through *de novo* synthesis [8]. The cell volume then restores progressively over time, and eventually, the cell adapts to the higher osmolarity. Intracellular crowding may act as a cell volume sensor to trigger osmoregulation [9–11]. Meanwhile, intracellular crowding due to volume reduction inevitably affects the cellular physiology globally, e.g., slowing down protein diffusion [12–18] and reducing the elongation speed of translating ribosomes [19, 20]. Despite extensive knowledge regarding the molecular details of osmotic response pathways [4], how intracellular crowding interferes with gene expression regulation and affects osmotic adaption remains an open question.

Interestingly, many features of microbial osmoresponses appear general across different organisms, suggesting a universal underlying mechanism. For example, it is widely observed that microbial cells can adapt to a broad range of external osmolarity, with the external osmotic pressure varying over an order of magnitude [19, 21–23]. Furthermore, the growth rate in the steady state decreases as the external osmolarity increases and a complete arrest of cell growth occurs above a critical osmolarity [19, 22–26]. Moreover, upon an osmotic shock, the growth rate usually does not approach the new steady-state value monotonically, e.g., an overshoot of growth rate often occurs upon a hypoosmotic shock [23], and a damped oscillation of growth rate can happen after a hyperosmotic shock [22]. In recent experiments of *Schizosaccharomyces pombe* [27], an oscillatory osmotic shock was applied to cells during which cell volume growth was dramatically slowed down while biomass was still actively produced. Surprisingly, a supergrowth phase happened after removing the oscillatory osmotic shock, during which cells grew much faster than the steady state before the shocks.

In this work, we unify all these phenomena by a theory capturing the essential elements of osmoresponses: physical constraints (e.g., the crowding effects and osmotic imbalance) and biological regulation, including osmoregulation (i.e., regulation of the osmolyte-producing protein) and cell-wall synthesis regulation. Our model assumes the following phenomenological rules: (1) the change in free water volume within the cell is driven by osmotic imbalance [7, 28], while the remaining volume changes in proportion to protein production; (2) osmoregulation influences the production of osmolyte-producing protein, governed by intracellular protein density [29]; (3) cell-wall synthesis is regulated through a feedback mechanism, wherein turgor pressure modulates the efficiency of cell-wall synthesis, enabling the cell to maintain a relatively stable turgor pressure; and (4) intracellular crowding slows down biochemical reactions as cytoplasmic density increases, with reactions ceasing entirely when protein density reaches a critical threshold. Upon a hyperosmotic shock, cell volume reduction due to water efflux increases the protein density, inducing the up-regulation of osmolyte-producing protein but slowing down the translation speed due to crowding. Upon a hypoosmotic shock, the dramatic water influx stretches the cell wall, and the increased turgor pressure induces cell-wall synthesis [5, 30, 31].

We remark that our model is coarse-grained, without including detailed molecular mechanisms, and is therefore applicable across diverse microbial species. Notably, the predicted steady-state growth rate as a function of internal osmotic pressure from our model aligns well with experimental data from diverse organisms. This alignment allows us to quantify the sensitivities of translation speed and regulation of osmolyte-producing protein in response to intracellular density. Additionally, we demonstrate that osmoregulation and cell-wall synthesis regulation enable cells to adapt to a wide range of external osmolarities and prevent plasmolysis. Our model also predicts a non-monotonic time dependence of growth rate and protein density as they approach steady-state values following a constant osmotic shock, in concert with experimental observations [22, 23]. Moreover, we show that a supergrowth phase can arise following a sudden decrease in external osmolarity, driven by cell-wall synthesis regulation, either through the direct application of a hypoosmotic shock or the withdrawal of an oscillatory stimulus. Remarkably, the predicted amplitudes of supergrowth (i.e., growth rate peaks) quantitatively agree with multiple independent experimental measurements.

In the following "Results" section, we begin by outlining the primary assumptions and equations of our model in the subsection "Model Description," which includes four parts, each addressing one of the four phenomenological rules. Additional details can be found in Methods. We then proceed to the subsection "Steady states in constant environments", where we employ our theoretical framework to analyze steady-state growth and examine how the growth rate varies with external osmolarity. In the "Transient dynamics after a constant osmotic shock" subsection, we investigate the time-dependent osmoresponse after a constant hyperosmotic and hypoosmotic shock. Finally, in "Comparison with experiments: supergrowth phenomena after osmotic oscillation", we address the supergrowth phenomena observed in *S. pombe*, utilizing our model to elucidate these experimental observations.

Results

Model Description

Cell growth

In the limit of an extreme hyperosmotic shock, the remaining cytoplasmic volume is comparable to the volume of expelled water [14, 21, 32]. Thus, the total cytoplasmic volume must be divided into a free volume and a bound volume [28, 33–36],

$$V = V_f + V_b. \tag{1}$$

The free volume comes from the free water that is osmotically active, and the bound volume includes the bound water V_{bw} (i.e., water of macromolecular hydration) and the volume of dry mass V_{bd} (**Figure 1A**). Because the fraction of protein mass in the total dry mass is typically constant and the volume of bound water is proportional to the dry mass [21], the bound volume is proportional to the total protein mass m_p through $V_b = \alpha m_p$. Here, α is a constant, and its values for some model organisms are included in **Table 1**, and its detailed calculations from experimental data are in Section B of Supplementary Material.

The free volume changes due to osmotic imbalance, and the growth rate of the free volume follows

$$\mu_f \equiv \frac{\dot{V}_f}{V_f} = k_w(\Pi_{in} - \Pi_{out} - \sigma), \tag{2}$$

where Π_{in}, Π_{out} are the internal (i.e., cytoplasmic) and external osmotic pressures, respectively [7]. Π_{in} is proportional to the concentration of osmolyte molecules in the free volume: $\Pi_{in} = k_B T N_a / V_f$ where N_a is the number of osmolyte molecules, k_B is the Boltzmann constant, and T is the temperature. For simplicity, we assume that the production speed of osmolyte molecules is proportional to the mass of osmolyte-producing protein (Methods). Here, we have replaced the

E. coli	**Value**	**Reference**		
σ_c	1 [atm]	Ref. [6]		
α	1.68 [ml/g]	Deduce from Ref. [32] (see Supplementary Material)		
	MBM [21]	**MOPS+fructose** [19]	**MOPS+glucose** [19]	**LB** [22]
$k_r^{\max}\chi_r$	0.743 [1/h]	0.776 [1/h]	1.14 [1/h]	2.05 [1/h]
$\Pi_{in,c}$	1.54 [Osm]	1.49 [Osm]	1.61 [Osm]	2.18 [Osm]
$H_r/(H_a+1)$	1.68	1.30	1.18	2.72

B. subtilis (LB)	**Value**	**Reference**
σ_c	19 [atm]	Ref. [33]
$k_r^{\max}\chi_r$	2.52 [1/h]	Fit to Ref. [23]
$\Pi_{in,c}$	3.09 [Osm]	
$H_r/(H_a+1)$	2.18	

S. cerevisiae (YPD)	**Value**	**Reference**
Π_{out}	0.26 [Osm]	
σ_c	3.1 [atm]	Ref. [35]
$\hat{\rho}_d$	0.295 [g/ml]	Ref. [43]
μ	0.448 [1/h]	Our experiment
f	0.6	Ref. [14]
ρ_p	0.155 [g/ml]	See Supplementary Material
ρ_c	0.994 [g/ml]	See Supplementary Material
α	4.29 [ml/g]	See Supplementary Material
$k_r^{\max}\chi_r$	0.450 [1/h]	Fit to our data
$\Pi_{in,c}$	3.52 [Osm]	
$H_r/(H_a+1)$	2.54	
H_r	3.03	Set according to $1-(\rho_p/\rho_c)^{H_r}=\mu/k_r^{\max}\chi_r$

S. pombe (YE5S)	**Value**	**Reference**
Π_{out}	0.2 [Osm]	Ref. [62]
σ_c	10 [atm]	Ref. [35]
$\hat{\rho}_d$	0.282 [g/ml]	Ref. [44]
ρ_p	0.104 [g/ml]	See Supplementary Material
μ	0.35 [1/h]	Ref. [27]
f	0.788	Fit to Ref. [18] (see Supplementary Material)
ϵ	0.584	Ref. [62]
α	2.60 [ml/g]	See Supplementary Material
G	17.1 [atm]	$G=\sigma/\epsilon$
$\Pi_{out,c}$	3.5 [Osm]	
k_w	100 [1/(min atm)]	
ρ_c	0.267 [g/ml]	Deduce from Ref. [18] (see Supplementary Material)
H_r	3.03	Copied from *S. cerevisiae* in YPD
H_a	0.974	Set according to $\Pi_{in}/\Pi_{in,c}=(\rho_p/\rho_c)^{H_a}$
$k_r^{\max}\chi_r$	0.371 [1/h]	Set according to $\mu_r=k_r^{\max}\chi_r\left(1-(\rho_p/\rho_c)^{H_r}\right)$
$k_BTk_a^{\max}\chi_a^{\max}$	2.25 [(atm ml)/(g min)]	Set according to $k_BTk_a^{\max}\chi_a^{\max}\eta_a\rho_p=k_r^{\max}\chi_r\Pi_{in}$
τ_{cw}^{-}	0.1 [min]	
τ_{cw}^{+}	12.5 [min]	Fit to Ref. [27]
H_{cw}	1.7	

TABLE 1.

Model parameters for different species in their corresponding reference growth media.

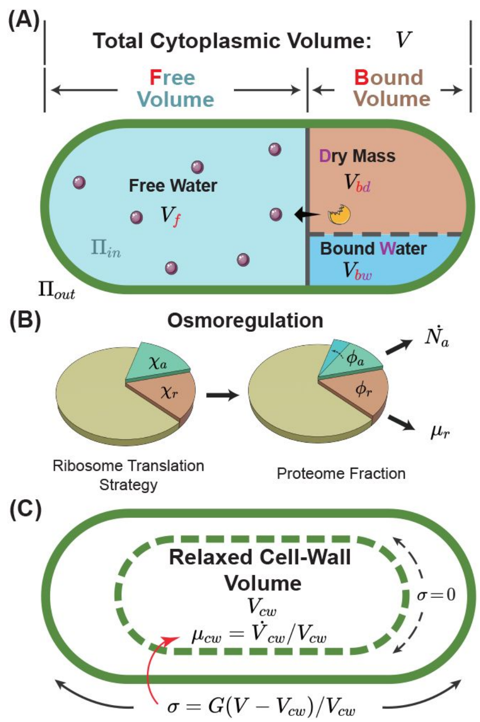

FIG. 1.

A schematic of the osmoresponse model.

(A) The total cytoplasmic volume includes the free and bound volumes. The free volume sets the internal osmotic pressure $\Pi_{in} = k_B T N_a / V_f$, where V_f is the free volume and N_a is the number of osmolyte molecules. The bound volume V_b comprises the dry mass V_{bd} and bound water V_{bw}, i.e., $V_b = V_{bd} + V_{bw}$, all proportional to the total protein mass. We model osmoregulation through the change of ribosome translation strategy. When the protein density increases, the fraction of ribosomes translating the osmolyte-producing protein χ_a is up-regulated, leading to the subsequent increase in the mass fraction of the osmolyte-producing protein Φ_a. The cell-wall synthesis process is controlled by the turgor pressure σ, which is proportional to the cell-wall strain $\epsilon = (V - V_{cw})/V_{cw}$. Here, V is the cytoplasmic volume, and V_{cw} is the relaxed cell wall volume.

difference of the hydrostatic pressures across the cell membrane with the turgor pressure σ, assuming that mechanical equilibrium is always satisfied. k_w is the filtration coefficient quantifying the water permeability of the cell membrane [37].

The species of osmolytes involved in osmoregulation are diverse across different microorganisms and conditions; nevertheless, they are primarily small organic molecules [8, 38]. In this work, we simplify the problem by considering a single species of osmolyte that dominates the internal osmotic pressure, e.g., glycerol in *Saccharomyces cerevisiae* [39–41] and glycine betaine in *Escherichia coli* [3], with the production speed proportional to the mass of the osmolyte-producing protein (**Figure 1A** and Methods).

To model gene expression regulation, we introduce χ_a and χ_r as the fractions of ribosomes translating the osmolyte-producing protein and ribosomal proteins (**Figure 1B** and Methods). In steady states, χ_a and χ_r are equal to the mass fractions of osmolyte-producing protein and ribosomal proteins in the total proteome, $\Phi_a = m_{p,a}/m_p$ and $\Phi_r = m_{p,r}/m_p$, respectively [29, 42]. In this work, we assume that the dry-mass growth rate is proportional to the fraction of ribosomal proteins within the total proteome for simplicity, $\mu_r = k_r m_{p,r}/m_p = k_r\Phi_r$. This assumption leverages the fact that ribosomes are responsible for producing all proteins. The proportionality coefficient k_r encapsulates the efficiency of ribosomal activity, being proportional to the elongation speed of the ribosome. We remark that k_r is influenced by the crowding effect, which we address later. The growth rate of the cytoplasmic volume, $\mu = \dot{V}/V$ is a weighted average of the free-volume growth rate μ_f and the dry-mass growth rate μ_r.

$$\mu = f\mu_f + (1-f)\mu_r. \tag{3}$$

Here, f is the free volume fraction in the total cytoplasmic volume: $f = V_f/V$. In this work, we refer to the growth rate as the growth rate of cytoplasmic volume μ unless otherwise mentioned.

Osmoregulation

Experiments found that the reduction of growth rate as the external osmolarity increases is dominated by the reduction of the translation speed k_r instead of the ribosomal fraction Φ_r [19]. Therefore, we assume that the fraction of ribosomes translating themselves χ_r is constant for simplicity. To model osmoregulation, we introduce a coupling between the fraction of ribosomes translating the osmolyte-producing protein χ_a and the degree of intracellular crowding. We quantify the crowding effects by the protein density, defined as $\rho_p = m_p/V_f$, which serves as a good proxy for the dry-mass density measured in the experiments [43, 44] (see **Table 1** and the detailed discussion on the relations between the two densities in Section A of Supplementary Material) and propose the following relation:

$$\chi_a = \chi_a^{\max}\left(\frac{\rho_p}{\rho_c}\right)^{H_a}. \tag{4}$$

Here, the parameter H_a quantifies the sensitivity of osmoregulation to intracellular crowding. ρ_c is the critical protein density above which intracellular processes are frozen, which we introduce later in **Eq. (8)**. Therefore, $\chi_a^{\max}$ represents the largest possible Φ_a since all intracellular dynamics is frozen when $\rho_p > \rho_c$. We remark that our model can be directly generalized to cases where osmolyte molecules are directly extracted from the environment. One only needs to change the interpretation of the parameter k_a in **Eq. (17)** from the synthesis rate to the uptake rate, and all the results are the same.

Cell-wall synthesis regulation

In this work, the cell wall is regarded as a linear elastic material, where the turgor pressure is proportional to the elastic strain of the cell wall by a constant modulus G such that

$$\sigma = G\epsilon = G\left(\frac{V}{V_{cw}} - 1\right). \quad (5)$$

Here, V_{cw} is the relaxed cell-wall volume (**Figure 1C**). When plasmolysis happens, the cell membrane detaches from the cell wall ($V < V_{cw}$), and the turgor pressure is zero. We introduce the growth rate of the relaxed cell-wall volume as $\mu_{cw} = \dot{V}_{cw}/V_{cw}$ .Given that in the steady states of cell growth, $\mu_r = \mu_{cw}$, we write μ_{cw} in the following form without loosing generality,

$$\mu_{cw} = \mu_r \eta_{cw}. \quad (6)$$

Here, η_{cw} is a coarse-grained parameter modeling the active regulation of cell-wall synthesis, which we refer to as the cell-wall synthesis efficiency in the following.

Experiments suggested that the turgor pressure induce cell-wall synthesis, e.g., through mechanosensors on cell membrane in *S. pombe* [45, 46], by increasing the pore size of the peptidoglycan network [5], and by accelerating the moving velocity of the cell-wall synthesis machinery in *E. coi* [31]. Guided by these ideas, we model the effects of turgor pressure on the time-dependence of the cell-wall synthesis efficiency as

$$\dot{\eta}_{cw} = \frac{1}{\tau_{cw}^{\pm}}\left[\left(\frac{\sigma}{\sigma_c}\right)^{H_{cw}} - \eta_{cw}\right]. \quad (7)$$

Here, σ_c is a characteristic scale of turgor pressure depending on species. τ_{cw}^{+} (τ_{cw}^{-}) is the relaxation timescale when the current η_{cw} is below (above) its target value $\eta_{cw}^{st} = (\sigma/\sigma_c)^{H_{cw}}$ .The former (latter) happens immediately after the cell is subject to a hypoosmotic (hyperosmotic) shock. In the extreme case of plasmolysis, the insertion of newly synthesized cell wall materials is interrupted immediately due to the separation of the cell membrane and cell wall. Meanwhile, the up-regulation of cell-wall synthesis rate presumably takes a longer time. For example, in fungi, where polarized growth is generally adopted, the up-regulation of the cell-wall synthesis rate involves reorienting the polarisome complex to the growing tip, directing actin polarization, and delivering cell-wall synthesis machinery [47, 48]. Therefore, we set $\tau_{cw}^{+} \gg \tau_{cw}^{-}$ in this work (see details of parameter values in **Table 1**).

Intracellular crowding

Multiple experiments suggested the cytoplasm of bacteria, yeast, and mammalian cells resemble crowded colloidal suspensions in which the mobilities of biomolecules are significantly reduced compared with dilute solutions [14, 15, 49–51], a signature of glass transition [52]. Intracellular crowding affects biochemical processes globally, e.g., slowing down translation and intracellular signaling by suppressing protein diffusion [14, 15, 18, 19, 49]. Therefore, the speed of osmolyte production, translational elongation, and cell-wall synthesis are all slowed down by the same crowding factor:

$$\eta_r = 1 - \left(\frac{\rho_p}{\rho_c}\right)^{H_r}. \quad (8)$$

Here, ρ_c is the critical protein density, and H_r is a parameter to quantify the sensitivity of biochemical reactions to the intracellular density. For example, the translational elongation speed is suppressed by intracellular crowding through $k_r = k_r^{max}\eta_r$ .Therefore, the dry-mass growth rate becomes $\mu_r = \mu_r^{max}\eta_r$,where we introduce $\mu_r^{max} = k_r^{max}\phi_r$.

The details of our model are summarized in Methods, with five independent variables: the protein density ρ_p, the mass fraction of osmolyte-producing protein Φ_a, the internal osmotic pressure Π_{in}, the cell-wall strain ϵ and the cell-wall synthesis efficiency η_{cw}. For convenience, Table S2 of the Supplementary Material provides a comprehensive list of all symbols used in the main text along with their meanings.

Steady states in constant environments

When cell growth reaches a steady state, the proportions of all components, including free water volume, cell mass, and cell wall volume, must be constant relative to the total cell volume to ensure homeostasis. Therefore, all growth rates in steady states of cell growth must be the same: $\mu_f = \mu_r = \mu_{cw}$. The consequence of cell-wall synthesis regulation can be seen directly from $\mu_{cw} = \mu_r$: the turgor pressure at steady states is constant, $\sigma = \sigma_c$. Experimentally, the cell-wall strain was measured by applying an acute hyperosmotic shock to induce plasmolysis, and it is approximately constant as the external osmolarity increases [22, 53], suggesting a constant turgor pressure independent of external osmolarity, in concert with our model assumptions. The internal osmotic pressure at steady states is related to the external osmotic pressure through Eq. (2),

$$\Pi_{in} = \Pi_{out} + \sigma. \qquad (9)$$

Here, we have neglected the term μ_f/k_w. Experimentally, an abrupt water flux occurs within hundreds of milliseconds after an osmotic shock [54], from which we can estimate the water permeability as $k_w \sim 100\ \mathrm{min}^{-1}\mathrm{atm}^{-1}$ considering an osmotic shock with an amplitude $\Delta\Pi_{out} = 1$ atm. Because the typical doubling times of microorganisms are from about 20 minutes to several hours, we estimate $\mu_f/k_w \sim 10-100$ Pa [54, 55], negligible compared with the typical cytoplasmic osmotic pressures, which can be several atmospheric pressures.

In steady states, the internal osmotic pressure is independent of time. Combining Eq. (4) and the dynamics of the internal osmotic pressure, Eq. (18c), we find the relationships between the protein density, the internal osmotic pressure, and the growth rate in the steady states:

$$\frac{\Pi_{in}}{\rho_p^{H_a+1}} = \mathrm{const}, \qquad (10a)$$

$$\frac{\mu_r}{\mu_r^{\max}} = 1 - \left(\frac{\Pi_{in}}{\Pi_{in,c}}\right)^{\frac{H_r}{H_a+1}}. \qquad (10b)$$

The right-hand side of Eq. (10a) is a constant independent of external osmolarity (see its detailed expression in Section C of Supplementary Material). In deriving Eq. (10b), we have replaced ρ_p by Π_{in} in Eq. (8) using Eq. (10a) with the critical internal osmotic pressure $\Pi_{in,c}$ proportional to ρ_c. Intriguingly, the relationship between the normalized growth rate $(\mu_r/\mu_r^{\max})$ and the normalized cytoplasmic osmotic pressure ($\Pi_{in}/\Pi_{in,c}$), which we refer to as the growth curve in the following, has only one parameter $H_r/(H_a + 1)$. Therefore, the growth curves of different organisms can be unified by a single formula, Eq. (10b), and different organisms may have different values of $H_r/(H_a + 1)$. Furthermore, Eq. (10b) predicts a critical external osmolarity $\Pi_{out,c} = \Pi_{in,c} - \sigma_c$, beyond which cell growth is completely inhibited.

We test the validity of Eq. (10b) by fitting it to the experimental growth curves (**Figure 2A**). To do this, we calculate the internal osmotic pressure using **Eq. (9)** given the values of the external osmotic pressure and the turgor pressure (**Table 1**). Intriguingly, the growth curves of multiple species can be well fitted by **Eq. (10b)**, from which we infer the parameters $H_r/(H_a + 1)$ and $\Pi_{in,c}$ (**Table 1**). We find that budding yeast cells exhibit a notable resilience to high external osmolarities: their $\Pi_{in,c}$ value is higher than those of Gram-positive bacteria, *B. subtilis*, and Gram-negative bacteria, *E. coli*. Further, budding yeast cells demonstrate a higher value of $H_r/(H_a + 1)$, indicating a reduced susceptibility to growth rate reduction when exposed to mild increases in the external osmolarity. Meanwhile, the osmoadaptation capability of *E. coli* depends on the growth media, presumably arising from variations in metabolic fluxes and gene expressions [19, 21, 22].

To further reveal the biological functions of biological regulations, we study the steady-state properties of mutant cells in which either osmoregulation or cell-wall synthesis regulation is depleted. For mutant cells with-out osmoregulation, $H_a = 0$ in Eq. (4). In this case, the fraction of osmolyte-producing protein is constant with time, i.e., $\phi_a = \chi_a^{\max}$.Comparing the dynamics of osmolyte and total protein mass, $\dot{N}_a = k_a\phi_a m_p$ and $\dot{m_p} = k_r\phi_r m_p$,one finds that the ratio of the number of osmolyte molecule and the total protein mass remains constant over time, irrespective of variations in external osmolarity (see the detailed derivation in Section C of Supplementary Material). As the external osmolarity increases, the protein density of mutant cells quickly reaches the critical value ρ_c according to Eq. (10a) with $H_a = 0$. Therefore, the steady-state growth curve of the mutant cells terminates at an external osmolarity much smaller than wild-type (WT) cells (**Figure 2B**), in agreement with previous experiments [56].

For mutant cells without the cell-wall synthesis regulation, $H_{cw} = 0$; therefore, the cell-wall synthesis efficiency η_{cw} equals 1 independent of time. Thus, the growth rate of the relaxed cell-wall volume is always equal to the growth rate of total protein mass [Eqs. (6, 7)]. Interestingly, in this case, the turgor pressure at steady states decreases with the increase of external osmolarity (**Figure 2C** and see the detailed proof in Section C of Supplementary Material). The decreased turgor pressure lowers the internal osmotic pressure given the same Π_{out} according to **Eq. (9)**, leading to a lower protein density of mutant cells than WT cells according to **Eq. (10a)**. Therefore, mutant cells grow faster than WT cells under the same external osmolarity (**Figure 2B**). Nevertheless, the mutant cells are prone to plasmolysis at a threshold external osmolarity where the WT cells can maintain constant turgor pressure (see the vertical line in **Figure 2C** around 2 M extra external osmolarity). Reduced turgor pressure is detrimental to multiple biological processes, e.g., cytokinesis in fission yeast requires the participation of turgor pressure [57].

To summarize, osmoregulation allows cells to grow in a wide range of external osmolarity conditions with a mild change in protein density. The cell-wall synthesis regulation allows cells to maintain a stable turgor pressure and avoid plasmolysis. Both regulatory mechanisms expand the range of external osmolarities that cells can adapt to.

Transient dynamics after a constant osmotic shock

Next, we study the dynamical behaviors of cellular properties in response to a constant osmotic shock: the external osmolarity changes abruptly and keeps its value for an infinitely long time. Intriguingly, we find that the dynamics of osmoresponse can be split into shock and adaptation periods (see insets of **Figure 3C, D**). The immediate water flow due to osmotic imbalance occurs in the shock period, during which the mass and osmolyte productions are negligible. Therefore,

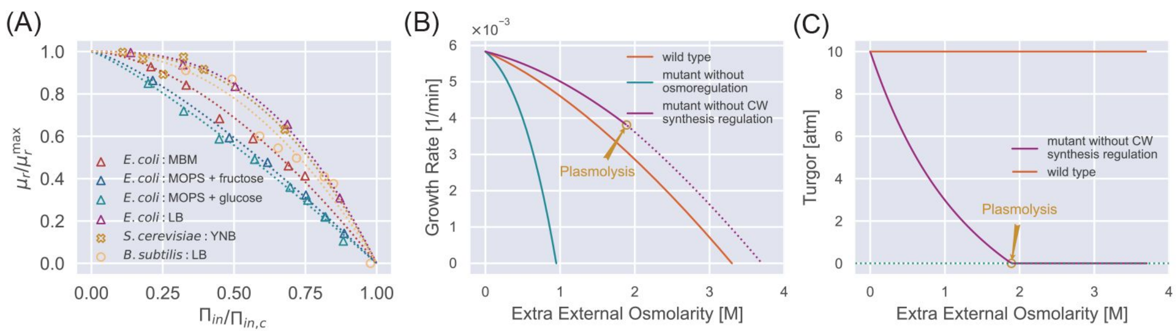

FIG. 2.

Steady-state properties under a constant external osmolarity.

(A) Normalized growth rate vs. normalized internal osmotic pressure of different species under various culture media. The experiment data (scatter markers) are fitted by our theoretical prediction Eq. (10b). The data of *E. coli* are from [19, 21, 22], the data of *B. subtilis* is from [23], and the data of *S. cerevisiae* is from our own experiments. (B) Growth curves of WT cells, mutant cells without osmoregulation ($H_a = 0$), and mutant cells without cell-wall synthesis regulation ($H_{cw} = 0$). The dotted line indicates the region where plasmolysis occurs for the mutant cells with $H_{cw} = 0$. (C) Mutant cells without cell-wall synthesis regulation cannot maintain a stable turgor pressure in a hypertonic environment, while WT cells can maintain a constant turgor pressure. The mutant cells reach plasmolysis at a threshold of external osmolarity. In (B) and (C), the parameters for WT cells are chosen as the values for *S. pombe*, and the mutant values are set such that they have the same growth rate as the WT cells in the reference medium (Table S2).

the ratio of the internal osmotic pressure and the protein density is invariant right before and after a shock period: $\Pi_{in}^i/\rho_p^i = \Pi_{in}^f/\rho_p^f$ where the upper index *i* (*f*) means the state right before (after) the shock period. Given this condition, we introduce the normalized protein density $\tilde{\rho}_p$ as

$$\tilde{\rho}_p = \frac{\rho_p}{\bar{\rho}_p}, \tag{11}$$

where the normalization factor $\bar{\rho}_p \propto \Pi_{in}$ (see its detailed expression in Methods) so that $\tilde{\rho}_p$ changes continuously across the shock period. Interestingly, we find that osmoresponse are governed by a two-dimensional dynamical system composed of $\tilde{\rho}_p$ and $\eta_a \equiv \phi_a/\chi_a^{\max}$ (Methods):

$$\frac{\dot{\tilde{\rho}}_p}{\tilde{\rho}_p} = \mu_r^{\max}\eta_r\left(1 - \tilde{\rho}_p\eta_a\right). \tag{12a}$$

$$\dot{\eta}_a = \mu_r^{\max}\eta_r\left[\left(\frac{\tilde{\rho}_p}{\tilde{\rho}_c}\right)^{H_a} - \eta_a\right], \tag{12b}$$

Here, $\tilde{\rho}_c = \rho_c/\bar{\rho}_p$ is the normalized critical protein density, and η_a denotes the efficiency of osmoregulation. From the above equations, it is clear that the timescale of osmoregulation is set by the doubling time: it takes about the doubling time for the protein density and the fraction of osmolyte-producing protein to adapt to the new steady-state values. For walled cells, $\tilde{\rho}_c$ and $\bar{\rho}_p$ depend on the time since $\Pi_{in} = \Pi_{out} + \sigma$ and the turgor pressure σ is time-dependent during osmoresponse processes (**Figure 3A**, B). For unwalled cells, such as mammalian cells and microbial cells with cell walls removed (i.e., protoplasts), $\tilde{\rho}_c$ is constant in a fixed environment (see detailed discussion on the transient dynamics of unwalled cells in Section D of Supplementary Material).

Upon a constant hyperosmotic shock, the immediate water efflux leads to an instantaneous drop in turgor pressure and a rise in protein density (**Figure 3A**). The internal state of the cell, $(\tilde{\rho}_p, \eta_a)$ evolves towards the new equilibrium point, $(\tilde{\rho}_c^{H_a/(H_a+1)}, \tilde{\rho}_c^{-H_a/(H_a+1)})$ One should note that the equilibrium point is time-dependent initially but eventually becomes fixed as the turgor pressure relaxes to the steady-state value (**Figure 3C** and Movie S1). Interestingly, the protein density increases initially and then decreases after the shock (**Figure 3A**). The decrease in protein density is because of the osmoregulation process, which is set by the doubling time [**Eqs. (12a**, **12b**)]. Meanwhile, we find that the initial increase of protein density is because of the suppressed growth of the relaxed cell-wall volume due to the low turgor pressure. Indeed, for unwalled cells, the protein density ρ_p decreases immediately after the shock (Figure S2B). We note that the growth rate approaches the new steady-state value non-monotonically (**Figure 3A**) because of the spiral trajectory in the space of the internal state (**Figure 3C**), consistent with experimental observations [22].

The phenomena are essentially the opposite for a constant hypoosmotic shock (**Figure 3B, D** and Movie S2). However, we find an extremely fast cell growth after the hypoosmotic shock, with a growth rate peak occurring about 25 minutes after applying the shock, which we call the supergrowth phase [27]. One should note that 25 minutes is much shorter than the doubling time (about two hours) but comparable to the timescale of cell-wall synthesis regulation, which we set as $\tau_{cw}^+ = 12.5$ min in the simulations in **Figure 3** (we will explain why we choose $\tau_{cw}^+ = 12.5$ min in the next section). Furthermore, applying a hypoosmotic shock to an unwalled cell does not induce a significant supergrowth phase compared with walled cells (Figure S2D).

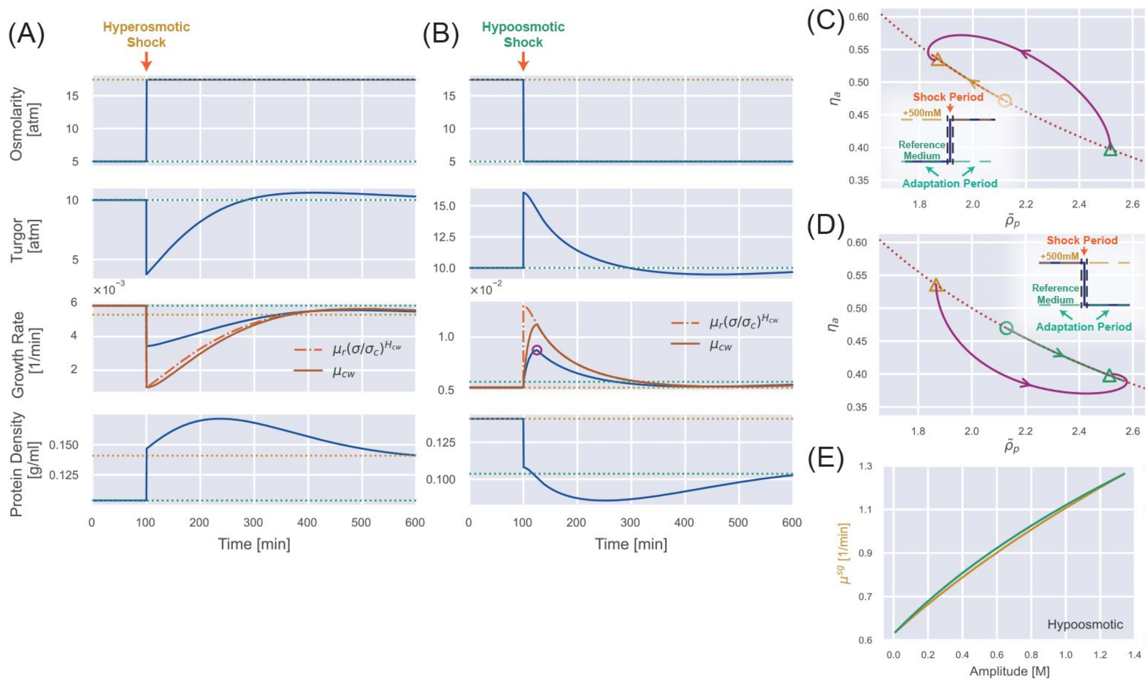

FIG. 3.

Transient dynamics after a constant osmotic shock.

(A) Numerical simulations of cells undergoing a constant 500 mM hyperosmotic shock. The dotted lines represent the steady-state values for the reference growth medium (green) and the medium after perturbation (yellow). (B) Numerical simulations of cells undergoing a constant 500 mM hypoosmotic shock. The purple circle in the third panel marks the growth rate peak during the supergrowth phase. (C) The dynamics of the internal state of a cell characterized by $(\tilde{\rho}_p, \eta_a)$.The dotted curve represents the constraint on the steady-state solution $\tilde{\rho}_p \eta_a = 1$,and the solid trajectory is from numerical simulations. The triangles indicate the steady-state solution before the perturbation and the steady-state solution after the perturbation for a long enough time. The yellow open circle represents the immediate steady-state solution after applying the hyperosmotic shock. (D) The same analysis as (C) but for a constant 500 mM hypoosmotic shock. (E) The growth rate peak in the supergrowth phase (yellow) and the immediate value of turgor pressure after the hypoosmotic shock σ^f (green) vs. the amplitude of the hypoosmotic shock.

We propose that supergrowth comes from the high turgor pressure caused by the hypoosmotic shock, which leads to a fast cell-wall synthesis according to **Eq. (7)**. Rapid insertion of materials into the cell wall relaxes the turgor pressure and allows the cells to grow faster [**Eqs. (2**, **3**)]. This idea is consistent with the observation that the growth rate and the growth rate of the relaxed cell-wall volume μ_{cw} reach their peaks simultaneously (**Figure 3B**). This observation also suggests that the timescale of supergrowth, i.e., the timing of growth rate peak, is set by the time scale of cell-wall synthesis regulation [τ_{cw}^{+} in **Eq. (7)**]. Notably, in the initial stage of the adaptation period, μ_{cw} approaches its target from below and reaches its target value at the growth rate peak (i.e., $\mu_r(\sigma/\sigma_c)^{H_{cw}}$) (the third panel of **Figure 3B**), after which μ_{cw} sticks to its target value and decreases accordingly because of the short relaxation time τ_{cw}^{-} [**Eq. (7)**]. A detailed proof of the conditions for supergrowth, including the necessity of a cell wall and the regulation of cell-wall synthesis, is provided in Section E of the Supplementary Material.

Following the discussion above, we obtain an analytical expression of the growth rate peak after a hypoosmotic shock (see the detailed derivations in Section F of Supplementary Material)

$$\mu^{sg} = \mu_r \left\{ 1 + \frac{f}{f + \frac{\Pi_{in}}{\sigma + G}} \times \left[\left(\frac{\sigma}{\sigma_c} \right)^{H_{cw}} - 1 \right] \right\}. \quad (13)$$

Here, all the variables on the right side are at the growth rate peak. Because the timescale of the osmoresponse process, which is around hours (**Figure 3B**), is much longer than the timescale of the supergrowth phase, which is about 20 minutes for *S. pombe* [27], the turgor pressure at the growth rate peak can be well approximated by its immediate value after the shock. Therefore, the growth rate peak must increase as the amplitude of the hypoosmotic shock increases, which we confirm numerically (**Figure 3E**).

Comparison with experiments: supergrowth phenomena after osmotic oscillation

Next, we quantitatively compare our theoretical predictions regarding the supergrowth phase with experimental data. In Ref. [27], Knapp et al. applied an osmotic oscillation to fission yeast *S. pombe* during which the external osmolarity alternated between two values.

They found cell growth was almost inhibited during the perturbation, while the protein and dry-mass densities increased. Surprisingly, cells grew unusually fast after the osmotic oscillation was removed and reached their maximum growth rate about 20 minutes after the end of the osmotic oscillation. The maximum growth rate can be twice the growth rate in the reference growth medium, and the elevation in growth rate can persist for 2-3 cell cycles. These observations are very similar to our results for a constant hypoosmotic shock (**Figure 3B**).

To test if our osmoresponse model captures the supergrowth phase for a periodic perturbation, we simulate a wide-type *S. pombe* cells with the same protocols as the experiments (see details of simulations in Methods). Intriguingly, our model successfully recapitulates the supergrowth phase and the gradually increasing protein density and dry-mass density during the perturbation (**Figure 4A**). We confirm that the cell-wall synthesis regulation is crucial for the emergence of the supergrowth phase since unwalled cells do not exhibit supergrowth under the same conditions (Figure S3B). Interestingly, we find that an infinitely long periodic osmotic shock can be equivalently mapped to a constant osmotic shock (see the detailed discussions and proof in Section D of Supplementary Material), which means that they have the same time-averaged growth rate and protein density in the steady states (Figure S2F).

In Ref. [27], the authors measured the growth rate peaks vs. three different parameters of the osmotic oscillations: amplitude, period length, and number of periods. We first fit the growth rate peaks vs. the amplitudes (**Figure 4D**), from which we obtain H_{cw} = 1.7, the sensitivity of the cell-

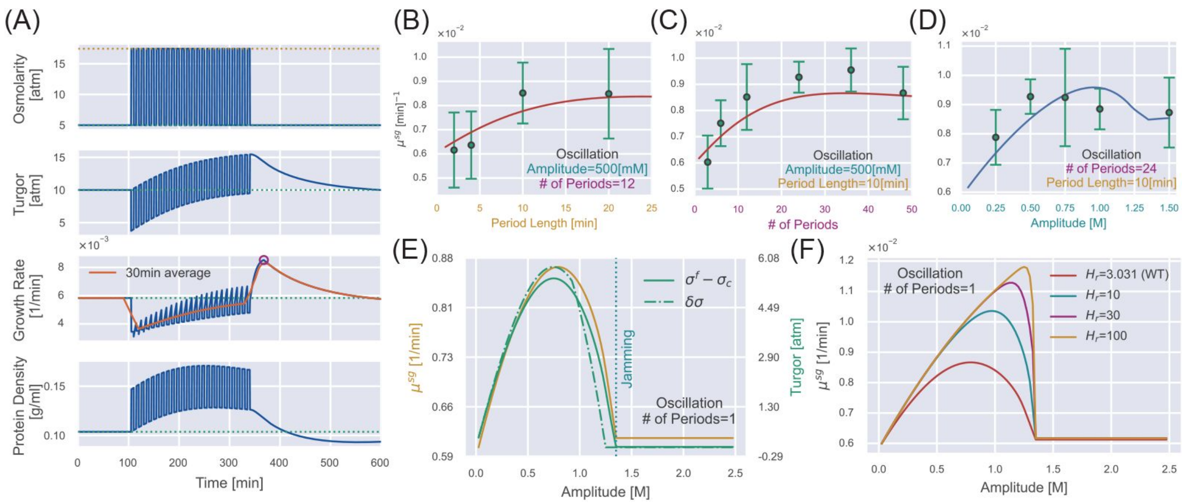

FIG. 4.

Supergrowth phenomena under osmotic oscillation.

(A) Numerical simulations of WT *S. pombe* undergoes 24 cycles of 500 mM osmotic oscillations with a 10-minute period. We show a 30-minute window average in the third panel of growth rate. (B-D) Quantitative agreement between simulations and experiments for the growth rate peak μ^{sg} vs. different oscillation parameters, including (B) amplitude, (C) period length, and (D) number of periods. The red lines in (B, C) are predictions, and the blue line in (D) is fitting from which we infer the values of H_{cw} and τ_{cw}^{+}.Green dots with error bars are experimental data from Ref. [27]. (E) In the case of osmotic oscillation with a single period, the hyperosmotic period persists for 120 min before reverting to the reference medium. The vertical dotted blue line represents the minimal amplitude to induce cytoplasm jamming during the hyperosmotic period. The excess turgor pressure $\sigma^f - \sigma_c$ upon exiting the hyperosmotic period is approximately equal to the recovered turgor pressure $\delta\sigma$ during the hyperosmotic period. (F) The growth rate peak μ^{sg} at different H_r vs. the amplitude of a single oscillation. H_r = 3.031 is the value of the WT *S. pombe*. Parameters of WT *S. pombe* are used in this figure unless otherwise mentioned (**Table 1**).

wall synthesis efficiency to turgor pressure [**Eq. (7)**], and $\tau_{cw}^{+} = 12.5$ min, the timescale in the up-regulation of cell-wall synthesis efficiency (which is why we set $\tau_{cw}^{+} = 12.5$ min in the previous section). Other model parameters are inferred from independent steady-state measurements, and we set the timescale in the down-regulation of cell-wall synthesis efficiency as $\tau_{cw}^{-} = 0.1$ min for simplicity (**Table 1**). We next fix the values of H_{cw} and τ_{cw}^{+} and plot the predicted growth rate peaks vs. the period length (**Figure 4B**) and number of periods (**Figure 4C**). As a strong support of our model, our predictions quantitatively match the experimental data without any further fitting.

Two interesting features of the curve μ^{sg} v.s amplitude catch our attention: the non-monotonic behavior and the kink point at which the derivative is discontinuous (**Figure 4D**), which are conserved regardless of the number of periods (Figure S7). Therefore, we study the case of a single oscillation for simplicity, which is equivalent to a hyperosmotic shock of finite duration. For a mild hyperosmotic shock, during the period of hyperosmotic shock, the turgor pressure has almost recovered to the steady-state value σ_c (**Figure 3A**). Therefore, switching from a long hyperosmotic period to the reference growth medium is equivalent to a constant hypoosmotic shock, where we have shown that the growth rate peak increases with the amplitude (**Figure 3E**). However, the crowding effect becomes more pronounced as the amplitude increases. Beyond the critical amplitude at the kink point, the cytoplasm is completely jammed during the hyperosmotic shock such that the cell states are precisely the same before and after the hyperosmotic shock, which means no supergrowth phase beyond this critical amplitude. Therefore, the curve μ^{sg} vs. amplitude must be non-monotonic (**Figure 4E**). Notably, for a very large H_r, cells can feel the crowding effect only when the cytoplasm is close enough to the critical protein density, shown as the abrupt decline of μ^{sg} (**Figure 4F**).

Finally, we remark that the significance of supergrowth is intimately related to the amount of recovered turgor pressure during the hyperosmotic shock $\delta\sigma$. We prove that the overshoot of turgor pressure after the removal of hyperosmotic shock ($\sigma^f - \sigma_c$), which sets the growth rate peak, is mainly set by the recovered turgor pressure during the hyperosmotic shock (see the detailed discussions in Section G of Supplementary Material). Indeed, μ^{sg}, $\sigma^f - \sigma_c$, and $\delta\sigma$ are highly correlated as we change the amplitude (**Figure 4E**).

Discussion

This study presents a theory of microbial osmoresponses based on a physical foundation and simplified biological regulation strategies. Our theory captures the steady-state properties of constant turgor pressure and reduced growth rate with increasing external osmolarity. We remark that the growth rate reduction is due to the loss of free water and subsequent intracellular crowding as the external osmolarity increases. In particular, we predict a critical external osmolarity above which cell growth is completely inhibited and a universal relationship between the normalized growth rate and the normalized internal osmotic pressure, fitting the data of bacteria and yeast. We also demonstrate the biological functions of osmoregulation and cell-wall synthesis regulation. Cells defective in osmoregulation cannot grow even if the external osmolarity is only mildly higher than the reference value. Cells defective in cell-wall synthesis regulation cannot maintain turgor pressure as the external osmolarity increases even though they grow faster than WT cells (**Figure 2B**), which will be a strong support of our theory if confirmed by experiments.

Regarding dynamic behaviors, our model predicts a non-monotonic time dependence of protein density after a constant hyperosmotic shock. We also unveil the supergrowth phase after a hypoosmotic shock, initially discovered in fission yeast after an osmotic oscillation [27]. As a strong support of our theory, the predicted growth rate peaks quantitatively agree with the experimental data without additional fitting. We demonstrate the critical role of cell-wall synthesis regulation in the supergrowth phenomenon (Section E of Supplementary Material). In Ref. [27], the authors observed the rapid repolarization of the cell-wall glucan synthase Bgs4 to the cell tip following the removal of osmotic oscillations in fission yeast, in agreement with the dynamics of the cell-wall synthesis efficiency predicted from our model (compare Figure S11 in this work and Figure S4H in Ref. [27]). To test our theories, we propose applying a hyperosmotic shock with a finite duration and measuring the growth rate after removing the hyperosmotic shock. We predict that the growth rate peak during the supergrowth phase is a nonmonotonic function of shock amplitude, initially rising because of the increased excess turgor pressure and later declining because the protein density reaches the critical value ρ_c during the shock (**Figure 4E**).

We remark that our model is intrinsically a coarsegrained model with many molecular details regarding gene expression regulation neglected, which allows us to gain more analytical insights. In Ref. [58], the authors studied the responses to osmotic stress in glucose-limited environments and found that cells exhibited stronger osmotic gene expression response under glucose-limited conditions than under glucose-rich conditions. Using a computational model based on molecular mechanisms combined with experimental measurements, the authors demonstrated that in a glucose-limited environment, glycolysis intermediates were limited, which required cells to express more glycerol-production enzymes for stress adaptation. In the current version of our model, we do not account for the interaction between cell growth and osmolyte production; instead, we assume a constant fraction of ribosomes dedicated to translating ribosomal proteins. Our model can be further generalized to include the more complex interactions, including the coupling between biomass and osmolyte production, e.g., by allowing the fraction of ribosomes translating (χ_r) to depend on the translation strategy of the osmolyte-producing enzyme (χ_a).

Ref. [22] showed that the expansion of *E. coli* cell wall is not directly regulated by turgor pressure, and this scenario is also included in our model as the case of $H_{cw} = 0$. According to our model, the supergrowth phase is absent if $H_{cw} = 0$ (Figure S8), consistent with the absence of a growth rate peak after a hypoosmotic shock in the experiments of *E. coli* [22]. Meanwhile, our predictions are consistent with the growth rate peak after a hypoosmotic shock observed for *B. subtilis* [23].

We remark several limitations of our current coarsegrained model. First, the high membrane tension that inhibits transmembrane flux of peptidoglycan precursors, leading to a growth inhibition before the supergrowth peak [23] is beyond our model. Second, in our current framework, the osmoregulation and cell-wall synthesis regulation rely on the instantaneous cellular states. However, microorganisms can exhibit memory effects to external stimuli by adapting to their temporal order of appearance [59]. Notably, in the osmoregulation of yeast, a short-term memory, facilitated by post-translational regulation of the trehalose metabolism pathway, and a longterm memory, orchestrated by transcription factors and mRNP granules, have been identified [60]. Besides, our model does not account for the role of osmolyte export in osmoregulation [61] and the interaction between biomass and osmolyte production [58]. Extending our model to include more realistic biological processes will be interesting.

In this work, we construct a systems-level and coarsegrained model capable of elucidating the complex processes underlying microbial osmoresponse. By leveraging the separation of timescales associated with mechanical equilibrium, cell-wall synthesis regulation, and osmoregulation, our model facilitates in-depth analytical and numerical analysis of how these various processes interact during cellular adaptation. In particular, we demonstrate the key physiological functions of osmoregulation and cell-wall synthesis regulation. We then apply this model to interpret the unusual phenomenon of supergrowth observed in fission yeast. This application addresses an essential challenge in experimental studies: exclusive knockout experiments can be difficult, and mechanistic interpretations of experimental observations are often lacking. Our theoretical framework offers a valuable tool for understanding such phenomena, contributing to the fundamental knowledge of microbial physiology and developing predictive models for microbial behavior under osmotic stress.

Methods

Details of the osmoresponse model

We define the fractions of osmolyte-producing protein and ribosomal proteins in the total proteome as $\Phi_a = m_{p,a}/m_p$ and $\Phi_r = m_{p,r}/m_p$, respectively. To model gene expression regulation, we introduce χ_a and χ_r as the fractions of ribosomes translating the osmolyteproducing protein and ribosomal proteins such that

$$\dot{m}_{p,r} = k_r \chi_r m_{p,r} \;\Rightarrow\; \dot{\phi}_r = k_r \phi_r (\chi_r - \phi_r), \tag{14}$$

$$\dot{m}_{p,a} = k_r \chi_a m_{p,r} \;\Rightarrow\; \dot{\phi}_a = k_r \phi_r (\chi_a - \phi_a), \tag{15}$$

$$\dot{m}_p = k_r m_{p,r} \qquad \Rightarrow\; \mu_r = k_r \phi_r. \tag{16}$$

Here, k_r is proportional to the elongation speed of ribosomes on mRNAs divided by the protein mass of a single ribosome, which is affected by the global crowding effect as $k_r = k_r^{\max}\eta_r$ .Here, μ_r is the growth rate of total protein mass, which is also the growth rate of dry mass and bound volume in our model since they are all proportional.

The osmolyte molecules are produced by the osmolyteproducing protein, with the rate given by

$$\dot{N}_a = k_a m_{p,a}, \tag{17}$$

where $k_a = k_a^{\max}\eta_r$ is the osmolyte production rate including the crowding factor and $m_{p,a}$ is the mass of osmolyte-producing protein. We summarize the dynamical equations involved in the osmoresponse model:

$$\dot{\rho}_p = (\mu_r - \mu_f)\rho_p \tag{18a}$$

$$\dot{\eta}_a = \mu_r \left[\left(\frac{\rho_p}{\rho_c} \right)^{H_a} - \eta_a \right] \tag{18b}$$

$$\dot{\Pi}_{in} = k_B T k_a^{\max} \eta_r \phi_a \rho_p - \mu_f \Pi_{in} \tag{18c}$$

$$\dot{\epsilon} = (\mu - \mu_{cw})(\epsilon + 1) \tag{18d}$$

$$\dot{\eta}_{cw} = \frac{1}{\tau_{cw}^{\pm}} \left[\left(\frac{\sigma}{\sigma_c} \right)^{H_{cw}} - \eta_{cw} \right]. \tag{18e}$$

To describe the osmoregulation process using a two-dimensional dynamical system, we introduce the normalized protein density as

$$\tilde{\rho}_p = \frac{k_B T k_a^{\max} \chi_a^{\max}}{\mu_r^{\max}} \frac{\rho_p}{\Pi_{in}} \equiv \frac{\rho_p}{\bar{\rho}_p}, \tag{19}$$

Combining Eq. (11) and Eq. (18a), we obtain the dynamical equation for $\tilde{\rho}_p$ as

$$\frac{\dot{\tilde{\rho}}_p}{\tilde{\rho}_p} = \mu_r^{\max}\eta_r\left(1 - \tilde{\rho}_p\eta_a\right). \tag{20}$$

Using Eq. (15), we obtain the equation for $\eta_a = \phi_a/\chi_a^{\max}$ as

$$\dot{\eta}_a = \mu_r^{\max}\eta_r\left[\left(\frac{\tilde{\rho}_p}{\tilde{\rho}_c}\right)^{H_a} - \eta_a\right], \tag{21}$$

where $\tilde{\rho}_c = \rho_c/\bar{\rho}_p$ .The unique equilibrium point for the internal state is

$$\left(\tilde{\rho}_p^{eq}, \eta_a^{eq}\right) = \left(\tilde{\rho}_c^{\frac{H_a}{H_a+1}}, \tilde{\rho}_c^{-\frac{H_a}{H_a+1}}\right). \tag{22}$$

Details of numerical simulations

We employ the LSODA algorithm with automatic stiffness detection and switching [63], implemented in SciPy [64], to solve Eq. (18). The parameters used for numerical simulations of walled cells are listed in **Table 1**.

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Additional files

Supplementary Material

Movie S1

Movie S2

Movie S3

Movie S4

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Reviewer #1 (Public review):

Summary:

A theoretical model for microbial osmoresponse was proposed. The model assumes simple phenomenological rules: (i) the change of free water volume in the cell due to osmotic imbalance based on pressure balance, (ii) Osmoregulation that assumes change of the proteome partitioning depending on the osmotic pressure that affects the osmolyte-producing protein production, (iii) The cell-wall synthesis regulation where the change of the turgor pressure to the cell-wall synthesis efficiency to go back to the target turgor pressure, (iv) Effect of Intracellular crowding assuming that the biochemical reactions slows down for more crowding and stops when the protein density (protein mass divided by free water volume) reaches a critical value. The parameter values were found in the literature or obtained by fitting to the experimental data. The authors compare the model behavior with various microorganismcs (E. coli, B. subtils, S. Cerevisiae, S. pombe), and successfully reproduced the overall trend (steady state behavior for many of them, dynamics for S. pombe). In addition, the model predicts non-trivial behavior such as the fast cell growth just after the hypoosmotic shock, which is consistent with experimental observation. The authors further make experimentally testable predictions regarding mutant behavior and transient dynamics.

The theory assumes simple mechanistic dependence between core variables without going into specific molecular mechanisms of regulations. The simplicity allows the theory to apply to different organisms by adjusting the time scales with parameters, and the model successfully explains broad classes of observed behaviours. Mathematically, the model provides analytical expressions of the parameter dependencies and an understanding of the dynamics through the phase space without being buried in the detail. This theory can serve as a base to discuss the universality and diversity of microbial osmoresponse.

The coarse-grained nature of the model is the strength of the model in terms of its generality. However, it does not consider various regulations at the molecular level. Hence, certain adaptation features are not considered in the current version of the model. The updated manuscript discusses the pros and cons of the current approach.

https://doi.org/10.7554/eLife.102858.2.sa2

Reviewer #2 (Public review):

Summary:

In this study, Ye et al. have developed a theoretical model of osmotic pressure adaptation by osmolyte production and wall synthesis.

Strengths:

They validate their model predictions of a rapid increase in growth rate on osmotic shock experimentally using fission yeast. The study has several interesting insights which are of interest to the wider community of cell size and mechanics.

Comments on revisions:

The authors have in the revised manuscript addressed the aspects of the writing that were unclear. , that are listed previously as major and minor comments. We believe the issues raised by this reviewer have been adequately addressed in the manuscript.

https://doi.org/10.7554/eLife.102858.2.sa1

Author response:

The following is the authors' response to the original reviews

List of major changes

(1) We have emphasized the assumptions underlying our modeling approach in the third paragraph of the Introduction section.

(2) We have included a new paragraph in the Discussion section to compare our model with a molecular mechanism-oriented model.

(3) We have included a new paragraph at the end of the Introduction section to outline the main content of each subsection in Results and the logical connections between them. Correspondingly, the chapter hierarchy and section titles have been adjusted.

(4) The Supplementary Material includes an additional table (Table S2) that provides detailed explanations of the symbols used in the model.

(5) We have included a new paragraph in the Introduction section to explicitly emphasize the phenomenological nature of our model and its broad applicability.

(6) In the Osmoregulation subsection, we have added a discussion on how our model can be directly generalized to scenarios involving the environmental uptake of osmolytes.

(7) We have included a more detailed examination of the limitations inherent in our modeling approach in the second last paragraph of the Discussion section.

(8) In the third last paragraph of the Discussion section, we have explicitly demonstrated that our model does not conflict with the observation that, in *E. coli*, cell wall synthesis is not directly regulated by the turgor pressure.

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *A theoretical model for microbial osmoresponse was proposed. The model assumes simple phenomenological rules: (i) the change of free water volume in the cell due to osmotic imbalance based on pressure balance, (ii) osmoregulation that assumes change of the proteome partitioning depending on the osmotic pressure that affects the osmolyte-producing protein production, (iii) the cell-wall synthesis regulation where the change of the turgor pressure to the cell-wall synthesis efficiency to go back to the target turgor pressure, (iv) effect of Intracellular crowding assuming that the biochemical reactions slow down for more crowding and stops when the protein density (protein mass divided by free water volume) reaches a critical value. The parameter values were found in the literature or obtained by fitting to the experimental data. The authors compare the model behavior with various microorganisms (*E. coli, B. subtils, S. Cerevisiae, S. pombe*), and successfully reproduced the overall trend (steady state behavior for many of them, dynamics for* S. pombe*). In addition, the model predicts non-trivial behavior such as the fast cell growth just after the hypoosmotic shock, which is*

> *consistent with experimental observation. The authors further make experimentally testable predictions regarding mutant behavior and transient dynamics.*
>
> *Strength:*
>
> *The theory assumes simple mechanistic dependence between core variables without going into specific molecular mechanisms of regulations. The simplicity allows the theory to apply to different organisms by adjusting the time scales with parameters, and the model successfully explains broad classes of observed behaviors. Mathematically, the model provides analytical expressions of the parameter dependences and an understanding of the dynamics through the phase space without being buried in the detail. This theory can serve as a base to discuss the universality and diversity of microbial osmoresponse.*

We would like to thank Reviewer 1 for thoroughly reading our work and appreciating our theoretical approach to investigating microbial osmotic response.

> *Weakness:*
>
> *The core part of this model is that everything is coupled with growth physiology, and, as far as I understand, the assumption (iv) (Eq. 8) that imposes the global reaction rate dependence on crowding plays a crucial role. I would think this is a strong and interesting assumption. However, the abstract or discussion does not discuss the importance of this assumption. In addition, the paper does not discuss gene regulation explicitly, and some comparison with a molecular mechanismoriented model may be beneficial to highlight the pros and cons of the current approach*

We thank Reviewer 1 for their very helpful feedback. We have significantly revised the manuscript as suggested by Reviewer 1. See the detailed answers in the following.

> ***Reviewer #1 (Recommendations for the authors)***
>
> *(1) Explicitly stating the assumption (iv) in the abstract and discussing its role would help readers understand.*

In the revised manuscript, we have significantly rewritten the third paragraph of the Introduction section to emphasize our key assumptions as suggested by Reviewer 1, including the relationship between global reaction rate and crowding:

"Our model assumes the following phenomenological rules: (1) the change in free water volume within the cell is driven by osmotic imbalance (Cadart et al., Nature Physics, 2019; Rollin et al., Elife, 2023), while the remaining volume changes in proportion to protein production; (2) osmoregulation influences the production of osmolyte-producing protein, governed by intracellular protein density (Scott et al., Science, 2010); (3) cell-wall synthesis is regulated through a feedback mechanism, wherein turgor pressure modulates the efficiency of cell-wall synthesis, enabling the cell to maintain a relatively stable turgor pressure; and (4) intracellular crowding slows down biochemical reactions as cytoplasmic density increases, with reactions ceasing entirely when protein density reaches a critical threshold."

We have also modified the abstract to mention the crowding effects explicitly. Additionally, we have added a few sentences in the first and second paragraphs of the Discussion section to emphasize the importance of crowding effects to our conclusions regarding the growth rate reduction in steady states and the non-monotonic dependence of the growth rate peak on the shock amplitude after a hyperosmotic shock.

> *(2) I found [Shen W , Gao Z, Chen K, Zhao A, Ouyang Q, Luo C. The regulatory mechanism of the yeast osmoresponse under different glucose concentrations.* Iscience*. 2023 Jan 20;26(1)], which discusses the medium glucose concentration dependence of the response, focused on the gene regulatory circuit and the metabolic flux. As far as I understood, this paper considers the effect of the reallocation of resources but not the mechanical part of the osmoresponse such as pressure explicitly. It will be interesting to discuss the pros and cons in comparison with such a model. In principle, I will not be surprised if the current model does not differentiate the different glucose concentrations much since it is a more coarse-grained model, and I don't think it is a problem, but it will be good to have an explicit discussion.*

We appreciate Reviewer 1's insightful comment regarding the work by Shen et al. (iScience, 2023), which elucidates the two distinct osmoresponse strategies in yeast. By quantifying Hog1 nuclear translocation dynamics and downstream protein expression, the study reveals that in a rich medium, cells can leverage surplus glycolytic products as defensive reserves, reallocating metabolic flux to facilitate rapid adaptation to osmotic changes. Conversely, limited glycolytic intermediates in low-glucose environments necessitate increased enzyme synthesis for osmotic adaptation.

The paper highlighted by Reviewer 1 studies yeast's adaptive strategies under two stresses—nutrient limitation and osmotic pressure and provides an important complement to our study.

In our simplified model, we did not include the interaction between cell growth and osmolyte production, assuming a constant fraction of ribosomes translating ribosomal proteins, supported by the experiments of *E. coli* (Dai et al., mBio, 2018). We remark that incorporating competitive dynamics for translational resources into our framework can be achieved by modifying the proportion of ribosomes translating themselves (X_r), from a constant to a function related to the translation strategy of the osmolyte-producing enzyme ((X_a).

In the revised manuscript, we have included a new discussion in the third paragraph of the Discussion section to compare our approach with the molecular mechanism-oriented model:

“We remark that our model is intrinsically a coarse-grained model with many molecular details regarding gene expression regulation neglected, which allows us to gain more analytical insights. In [Shen et al., iScience, 2023], the authors studied the responses to osmotic stress in glucose-limited environments and found that cells exhibited stronger osmotic gene expression response under glucose-limited conditions than under glucose-rich conditions. Using a computational model based on molecular mechanisms combined with experimental measurements, the authors demonstrated that in a glucose-limited environment, glycolysis intermediates were limited, which required cells to express more glycerol-production enzymes for stress adaptation. In the current version of our model, we do not account for the interaction between cell growth and osmolyte production; instead, we assume a constant fraction of ribosomes dedicated to translating ribosomal proteins. Our model can be further generalized to include the more complex interactions, including the coupling between biomass and osmolyte production, e.g., by allowing the fraction of ribosomes translating ((X_r) to depend on the translation strategy of the osmolyte-producing enzyme ((X_a).”

> *(3) A minor comment: The authors call assumption (iii) (eq. 7) "positive feedback from turgor pressure to the cell-wall synthesis efficiency" (line 204). I have a hard time seeing this as positive feedback. It regulates the cell wall synthesis so that turgor pressure returns to the desired value; hence, isn't it negative feedback?*

We apologize for this confusion. We have removed the term "positive feedback" in the revised manuscript.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *In this study, Ye et al. have developed a theoretical model of osmotic pressure adaptation by osmolyte production and wall synthesis.*
>
> *Strengths:*
>
> *They validate their model predictions of a rapid increase in growth rate on osmotic shock experimentally using fission yeast. The study has several interesting insights which are of interest to the wider community of cell size and mechanics.*
>
> *Weaknesses:*
>
> *Multiple aspects of this manuscript require addressing, in terms of clarity and consistency with previous literature. The specifics are listed as major and minor comments.*
>
> *Major comments:*
>
> *(1) The motivation for the work is weak and needs more clarity.*

We thank Reviewer 2 for this very helpful comment, which we believe has significantly improved our manuscript. We would like to clarify the two major motivations of our study.

First, we aim to construct a systems-level and coarse-grained model capable of elucidating the complex processes underlying microbial osmoresponse. By leveraging the separation of timescales associated with mechanical equilibrium, cell-wall synthesis regulation, and osmoregulation, our model facilitates in-depth analytical and numerical analysis of how these various processes interact during cellular adaptation. In particular, we demonstrate the key physiological functions of osmoregulation and cell-wall synthesis regulation.

Second, we seek to apply this model to interpret the phenomenon of supergrowth observed in fission yeast *Schizosaccharomyces pombe* (Knapp et al., Cell Systems, 2019). This application addresses an essential challenge in experimental studies: exclusive knockout experiments can be difficult, and mechanistic interpretations of experimental observations are often lacking. Our theoretical framework offers a valuable tool for understanding such phenomena, contributing to the fundamental knowledge of microbial physiology and developing predictive models for microbial behavior under osmotic stress.

In the revised manuscript, we have included a new paragraph at the end of the Discussion section to emphasize our motivations better:

“In this work, we construct a systems-level and coarse-grained model capable of elucidating the complex processes underlying microbial osmoresponse. By leveraging the separation of timescales associated with mechanical equilibrium, cell-wall synthesis regulation, and osmoregulation, our model facilitates in-depth analytical and numerical analysis of how these various processes interact during cellular adaptation. In particular, we demonstrate the key physiological functions of osmoregulation and cell-wall synthesis regulation. We then apply this model to interpret the unusual phenomenon of supergrowth observed in fission yeast. This application addresses an essential challenge in experimental studies: exclusive knockout experiments can be difficult, and mechanistic interpretations of experimental observations are often lacking. Our theoretical framework offers a valuable tool for understanding such phenomena, contributing to the fundamental knowledge of microbial physiology and developing predictive models for microbial behavior under osmotic stress."

> *(2) The link between sections is very frequently missing. The authors directly address the problem that they are trying to solve without any motivation in the results section.*

We are grateful to Reviewer 2 for their valuable feedback. In the revised manuscript, we have included a new paragraph at the end of the Introduction section to outline the main content of each subsection in Results and the logical connections between them:

"In the following "Results" section, we begin by outlining the primary assumptions and equations of our model in the subsection "Model Description," which includes four parts, each addressing one of the four phenomenological rules. Additional details can be found in Methods. We then proceed to the subsection "Steady states in constant environments", where we employ our theoretical framework to analyze steady-state growth and examine how the growth rate varies with external osmolarity. In the "Transient dynamics after a constant osmotic shock" subsection, we investigate the time-dependent osmoresponse after a constant hyperosmotic and hypoosmotic shock. Finally, in "Comparison with experiments: supergrowth phenomena after osmotic oscillation", we address the supergrowth phenomena observed in *S. pombe*, utilizing our model to elucidate these experimental observations."

> *(3) The parameters used in the models (symbols) need to be explained better to make the paper more readable.*

We apologize for this confusion. In the revised Supplementary Material, we have included an additional table (Table S2) to explain the meanings of the symbols employed in the model to help the reader better understand.

> *(4) Throughout the paper, the authors keep switching between organisms that they are modelling. There needs to be some consistency in this aspect where they mention what organism they are trying to model, since some assumptions that they make may not be valid for both yeast as well as bacteria.*

We thank Reviewer 2 for this very helpful comment. We would like to clarify that our model is coarse-grained without including detailed molecular mechanisms; therefore, it presumably applies to various species of microorganisms. Indeed, the predicted steady-state growth curves derived from our model and the experimental data obtained from various organisms agree reasonably well (Figure 2A of the main text).

In the revised manuscript, we have explicitly emphasized the nature of our phenomenological model and its broad applicability in the fourth paragraph of the Introduction section:

"We remark that our model is coarse-grained, without including detailed molecular mechanisms, and is therefore applicable across diverse microbial species. Notably, the predicted steady-state growth rate as a function of internal osmotic pressure from our model aligns well with experimental data from diverse organisms. This alignment allows us to quantify the sensitivities of translation speed and regulation of osmolyte-producing protein in response to intracellular density. Additionally, we demonstrate that osmoregulation and cellwall synthesis regulation enable cells to adapt to a wide range of external osmolarities and prevent plasmolysis. Our model also predicts a non-monotonic time dependence of growth rate and protein density as they approach steady-state values following a constant osmotic shock, in concert with experimental observations (Rojas et al., PNAS, 2014; Rojas et al., Cell systems, 2017). Moreover, we show that a supergrowth phase can arise following a sudden decrease in external osmolarity, driven by cell-wall synthesis regulation, either through the direct application of a hypoosmotic shock or the withdrawal of an oscillatory stimulus. Remarkably, the predicted amplitudes of supergrowth (i.e., growth rate peaks) quantitatively agree with multiple independent experimental measurements.”

Furthermore, we have also included a comparison with a detailed molecular mechanism model in the third paragraph of the Discussion section:

“We remark that our model is intrinsically a coarse-grained model with many molecular details regarding gene expression regulation neglected, which allows us to gain more analytical insights. In [Shen et al., iScience, 2023], the authors studied the responses to osmotic stress in glucose-limited environments and found that cells exhibited stronger osmotic gene expression response under glucose-limited conditions than under glucose-rich conditions. Using a computational model based on molecular mechanisms combined with experimental measurements, the authors demonstrated that in a glucose-limited environment, glycolysis intermediates were limited, which required cells to express more glycerol-production enzymes for stress adaptation. In the current version of our model, we do not account for the interaction between cell growth and osmolyte production; instead, we assume a constant fraction of ribosomes dedicated to translating ribosomal proteins. Our model can be further generalized to include the more complex interactions, including the coupling between biomass and osmolyte production, e.g., by allowing the fraction of ribosomes translating (($X_{r</supb}$) to depend on the translation strategy of the osmolyte-producing enzyme ((X_a).”

> *(5) The extent of universality of osmoregulation i.e the limitations are not very well highlighted.*

The osmoregulation mechanism described in our model primarily addresses changes in cytoplasmic osmolarity through the de-novo synthesis of compatible solutes, widely observed across bacteria, archaea, and eukaryotic microorganisms. This review article (GundeCimerman et al., FEMS microbiology reviews, 2018) provides an extensive summary and exploration of the primary compatible solutes utilized by organisms from all three domains of life, underscoring the prevalence of this osmoregulatory strategy. Furthermore, our model can be directly generalized to scenarios involving the direct uptake of osmolytes from the environment. One only needs to change the interpretation of the parameter, k_a in the production of osmolyte molecule, $\dot{N}_a = k_a m_{p,a}$, from the synthesis rate to the uptake rate, and all the results are equally applicable. In the revised manuscript, we have briefly discussed this point in the subsection “Osmoregulation.”

We agree with Reviewer 2 that our model's coarse-grained nature makes it broadly applicable to diverse microbial taxa; however, more specialized adaptations are beyond our model. In the revised manuscript, we have included a more detailed examination of the limitations inherent in our modeling approach in the second last paragraph of the Discussion section:

“We remark several limitations of our current coarse-grained model. First, the high membrane tension that inhibits transmembrane flux of peptidoglycan precursors, leading to a growth inhibition before the supergrowth peak (Rojas et al., Cell systems 2017) is beyond our model. Second, in our current framework, the osmoregulation and cell-wall synthesis regulation rely on the instantaneous cellular states. However, microorganisms can exhibit memory effects to external stimuli by adapting to their temporal order of appearance (Mitchell et al., Nature 2009). Notably, in the osmoregulation of yeast, a short-term memory, facilitated by post-translational regulation of the trehalose metabolism pathway, and a long-term memory, orchestrated by transcription factors and mRNP granules, have been identified (Jiang et al., Science signaling 2020). Besides, our model does not account for the role of osmolyte export in osmoregulation (Tamas et al., Molecular microbiology, 1999) and the interaction between biomass and osmolyte production (Shen et al., Iscience 2023). Extending our model to include more realistic biological processes will be interesting.”

> *(6) Line 198-200: It is not clear in the text what organisms the authors are writing about here. "Experiments suggested that the turgor pressure induce cell-wall synthesis, e.g., through mechanosensors on cell membrane [45, 46], by increasing the pore size of the peptidoglycan network [5], and by accelerating the moving velocity of the cell-wall synthesis machinery [31]". This however is untrue for bacteria as shown by the study (reference 22 is this paper: E. Rojas, J. A. Theriot, and K. C. Huang, Response of escherichia coli growth rate to osmotic shock, Proceedings of the National Academy of Sciences 111, 7807 (2014).*

We thank Reviewer 2 for pointing out this very important issue and apologize for the confusion. References 45 and 46 (Dupres et al., Nature Chemical Biology 2009; Neeli-Venkata et al., Developmental Cell 2021) discuss how Wsc1 acts as a mechanosensor in *S. pombe*, detecting turgor pressure and activating pathways that reinforce the cell wall. Reference 5 (Typas et al., Cell 2010) explains the role of LpoA and LpoB, the two outer membrane lipoprotein regulators in *E. coli*, which modulate peptidoglycan synthesis in an extracellular manner. Reference 31 (Amir and Nelson, PNAS 2012) is a theoretical paper showing that turgor pressure may accelerate the moving velocity of the cell wall synthesis machinery in *E. coli*. In the revised manuscript, we have been more explicit about the organisms we refer to in the subsection “Cell-wall synthesis regulation.”

Meanwhile, we agree with Reviewer 2 that cell wall synthesis may not be directly regulated by turgor pressure in *E. coli* (Rojas et al., PNAS 2014). We would like to clarify that this scenario is also included in our model corresponding to $H_{cw} = 0$ (Eq. (7) in the main text): the turgor pressure does not affect the cell-wall synthesis. Therefore, the supergrowth phenomenon observed in *S. pombe* does not manifest under hypotonic stimulation in *E. coli.*

In the revised manuscript, we have emphasized this point more explicitly in the third last paragraph of the Discussion section:

“Reference 22 (Rojas et al., PNAS, 2014) showed that the expansion of E. *coli* cell wall is not directly regulated by turgor pressure, and this scenario is also included in our model as the case of $H_{cw} = 0$. According to our model, the supergrowth phase is absent if $H_{cw} = 0$ (Figure S8), consistent with the absence of a growth rate peak after a hypoosmotic shock in the experiments of *E. coli* (Rojas et al., PNAS, 2014). Meanwhile, our predictions are consistent with the growth rate peak after a hypoosmotic shock observed for *B. subtilis* (Rojas et al., Cell systems, 2017).”

> *(7) The time scale of reactions to hyperosmotic shocks does not agree with previous literature (reference 22). Therefore defining which organism you are looking at is important. Hence the statement " Because the timescale of the osmoresponse process, which is around hours (Figure 3B), is much longer than the timescale of the supergrowth phase, which is about 20 minutes, the turgor pressure at the growth rate peak can be well approximated by its immediate value after the shock." from line 447 does not seem to make sense. The authors need to address this.*

We apologize for this confusion. In the revised manuscript, we have clarified that the cited time scales are for the fission yeast *S. pombe* after Eq. (13) in the main text.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) Inconsistency in nomenclature: On line 117, the equation reads $V_b = am_p$ where V_b is the bound volume. Whereas bound volume has been referred to as V_{bd} previously and in*

> *Figure 1.*

Answer: We apologize for this confusion. In our model, the total bound volume V_b comprises the volume of dry mass and bound water, $V_b = V_{bd} + V_{bw}$, where V_{bd} is the volume occupied by dry mass and V_{bw} is the volume of bound water. In the revised manuscript, we have added a brief discussion of this point in the caption of Figure 1.

> *(2) Line 180: Please define $\varrho\varrho$ for equation 4.*

We apologize for this confusion. In the text, the symbol ϱ_p denotes the mass of a given substance per unit volume of free water, and its unit is g/ml. The specific substance in consideration is indicated by a subscript. For example, ϱ_p in Eq. (4) represents the protein density, and ϱ_c stands for the critical protein density, above which intracellular chemical reactions cease according to Eq. (8) of the main text. In the revised manuscript, we have clarified the meaning of ϱ_c after Eq. (4).

> *(3) Line 187: Equation 5 also needs to be explained better. Hence there is a need to be more specific while stating the assumptions.*

The elastic modulus G defined in Eq. (5) of the main text is a measure of the cell wall's resistance to volume expansion. We assume a constant G for simplicity, which is reasonable when the cell wall deformation is mild. In the revised manuscript, we have been more explicit about our assumptions regarding the turgor pressure in the subsection “Cell-wall synthesis regulation.”

> *(4) Line 225: For a biological audience some elaboration on "glass transition" may be required- either as a reference to a review or to a 1 sentence statement of relevance.*

We appreciate Reviewer 2’s helpful comment. In the revised manuscript, we have added a brief introduction to the glass transition and a citation to a review paper (Hunter and Weeks, Rep. Prog. Phys. 2012) at the beginning of the subsection “Intracellular crowding.”

> *(5) Line 247: "All growth rates in steady states of cell growth are the same: $\mu_f = \mu_r = \mu_{cw}$". The authors need to explain in a line or two why this is true. Since the processes are independent, it is safe to assume that all μ's are constant, but it is not obvious why they should all be equal.*

We apologize for the lack of a clear explanation regarding the equality of steady-state growth rates in our previous manuscript. In the revised manuscript, we have added a brief explanation of the equality of the three growth rates at the beginning of the subsection “Steady states in constant environments”:

“When cell growth reaches a steady state, the proportions of all components, including free water volume, cell mass, and cell wall volume, must be constant relative to the total cell volume to ensure homeostasis. Therefore, all growth rates in steady states of cell growth must be the same: $\mu_f = \mu_r = \mu_{cw}$.”

> *(6) Line 264: "Because the typical doubling times of microorganisms are around hours, we can estimate $\mu_f/k_w \sim 10$ Pa [51, 52] ..." since the authors are generalizing for yeast and bacteria, specifically E. coli, this is not a valid assumption to make. There is also a need to explain the basis of "$\mu_f/k_w \sim 10$ Pa".*

We appreciate the need for clarity in the estimation and its implications. The rough estimation of $\mu_f/k_w \sim 10$ Pa in the main text is given by:

$$\frac{\mu_f}{k_w} = \frac{100^{-1}[\min]^{-1}}{100\,[\min]^{-1}[\mathrm{atm}]^{-1}} = 10^{-4}[\mathrm{atm}] = 10[\mathrm{Pa}]. \tag{1}$$

Here, the typical value of μ_f (which equals to μ_r in steady state) is approximated by the inverse of the cell cycle, which is around hours. The estimation above is employed to justify the assumption that μ_f/k_w is much smaller than the cytoplasmic osmotic and turgor pressures, which can be several atmospheric pressures.

For the case of *E. coli*, based on the experimental results from Boer et al. (Boer et al., Biochemistry 2011), an 800mM hypoosmotic shock leads to a rapid expansion of cell volume accomplished within a time scale of 0.1s, from which we obtain:

$$\frac{\mu_f}{k_w} = \frac{20^{-1}[\min]^{-1}}{\frac{0.1^{-1}[s]^{-1}}{0.8 * 24.93[\mathrm{atm}]}} = 1.66 * 10^{-3}[\mathrm{atm}].$$

Therefore, our assumption that μ_f/k_w is much smaller than the cytoplasmic osmotic and turgor pressures is still valid.

In the revised manuscript, we have increased the estimation ranges to include the case of *E. coli* in the first paragraph of the subsection "Steady states in constant environments."

> *(7) Lines 279-283 need to be explained better.*

We apologize for the confusion. In the revised manuscript, we have explained more explicitly the meaning of the growth curve in the second paragraph of the subsection "Steady states in constant environments":

"Intriguingly, the relationship between the normalized growth rate (μ_r/μ_r^{max}) and the normalized cytoplasmic osmotic pressure ($\Pi_{in}/\Pi_{in,c}$), which we refer to as the growth curve in the following, has only one parameter $H_r/(H_a)$. Therefore, the growth curves of different organisms can be unified by a single formula, Eq. (10b), and different organisms may have different values of $H_r/(H_a + 1)$."

> *(8) In Figure 3, an arrow representing the onset of osmotic shock would make the figure more intuitive to understand.*

We appreciate Reviewer 2 for this helpful suggestion. We have modified Figure 3 as suggested.

> *(9) It is unclear to me if the growth rate μ_r is representative of the growth of total protein. This can be motivated better.*

We would like to clarify that the growth rate μ_r is defined as the changing rate of total protein mass divided by the total protein mass:

$$\mu_r \equiv \frac{\dot{m}_p}{m_p} = \frac{k_r m_{p,r}}{m_p} = k_r \phi_r. \tag{2}$$

Here, $m_{p,r}$ is the total mass of ribosomal proteins and $kkrr$ is a constant proportional to the elongation speed of ribosome. The expression of μ_r is a direct consequence of ribosomes being responsible for producing all proteins. In the revised manuscript, we have added more details in the introduction of the variable μ_r in the last paragraph of the subsection “Cell growth”:

“In this work, we assume that the dry-mass growth rate is proportional to the fraction of ribosomal proteins within the total proteome for simplicity, $\mu_r = k_r m_{p,r}/m_p = k_r \phi_r$. This assumption leverages the fact that ribosomes are responsible for producing all proteins. The proportionality coefficient k_r encapsulates the efficiency of ribosomal activity, being proportional to the elongation speed of the ribosome. We remark that $kkrr$ is influenced by the crowding effect, which we address later.”

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